

OMNIKON HACKATHON SUBMISSION



SafeGuard AI

Proactive Child Online Safety & Digital Trust Layer

THEME

Cyber Security, Blockchain & Digital Trust

CORE MISSION

Stateful behavioral risk modeling to protect youth from manipulation, grooming, and cyberbullying before harm escalates.

TEAM NAME

Cognitive Crew

TEAM MEMBERS

- **Niveditha B N**
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1. The Problem Statement

The Reactive Moderation Trap

Modern moderation architectures wait for direct user reports before triggering review workflows:

Harm Occurs → Child Reports → Platform Reviews → Action Taken

- Interventions take place after psychological or privacy harm has already manifested.
- Victims (especially young children) are often reluctant or hesitant to self-report.
- Fails to preempt rapidly escalating coordinated harassment.

Context Blindness & Keyword Flaws

Static string matching and isolated single-message filters fail fundamentally:

- **High False Positives:** Harmless peer banter or gaming phrases (e.g., "*I'll destroy you in this match*") trigger unnecessary account restrictions.
- **High False Negatives:** Multi-turn manipulation, coercion, and grooming bypass keyword filters by using innocuous, polite vocabulary.
- Inability to trace relationship dynamics across conversational turns.

The Digital Trust & Privacy Paradox

Balancing effective protection without violating fundamental digital privacy rights:

- **Over-Surveillance:** Heavy-handed parental spy tools erode child autonomy and breach trust.
- **Under-Protection:** Zero oversight leaves minors vulnerable to PII harvesting, financial scams, and predation.
- **Data Exposure:** Persistent storage of sensitive child chat transcripts creates high-value honeypots for data breaches.

2. Our Proposed Solution: SafeGuard AI

Stateful Behavioral Risk Engine

Shifts focus from *"Is this message toxic?"* to **"Is this interaction exhibiting an escalating risk pattern?"**

- Tracks multi-turn behavioral trajectories (Trust Building → Secrecy → PII Request → Coercion).
- Maintains an ephemeral conversation state vector rather than analyzing messages in isolation.
- Distinguishes healthy peer interaction from targeted asymmetric abuse.

Proportional & Age-Aware Interventions

Tiered intervention matrix designed to empower children and preserve conversational flow:

- **Low Risk** Silent passthrough; zero interruption of normal chats.
- **Medium Risk** Non-intrusive in-situ safety nudges & educational prompts.
- **Critical Risk** Message blurring, reporting guidance, trusted-adult escalation.

Privacy-by-Design & Digital Trust Architecture

Built to uphold child safety without compromising fundamental privacy standards:

01 Ephemeral Memory Window: Chats are evaluated within a transient sliding buffer with automated TTL purge.

02 Zero Raw-Chat Storage: Only anonymized risk indicators and metadata are transmitted to oversight dashboards.

03 PII Guardrails: Proactively alerts children when sensitive identifiers (phone, school, location, passwords) are solicited.

04 Explainable AI (XAI): Provides clear, human-interpretable reasoning factors for moderators and trusted adults.

3. Technical Architecture & Methodology

⚡ 2-Tier Inference Pipeline

Solves latency bottlenecks for real-time messaging ecosystems:

Tier 1: Fast Screening

Lightweight transformer & regex scoring (<30ms).
Filters 85%+ benign chats instantly.

↓ (If Anomaly > Threshold)

Tier 2: Contextual LLM Engine

Deep reasoning over conversation history to detect coercion, grooming, and power imbalances.

📐 Dynamic Risk Formulation

Mathematically models risk acceleration using Exponential Moving Average (EMA):

$$R_t = \lambda R_{t-1} + (1 - \lambda)(w_c C_t + w_b B_t + w_p P_t)$$

- R_t : Current dynamic conversation risk index.
- λ : Temporal memory decay factor (0.7).
- C_t : Semantic toxicity & hostility score.
- B_t : Behavioral anomaly vector (secrecy, isolation).
- P_t : Sensitive PII solicitation score.

💻 Technology Stack

Modular, production-ready engineering stack:

- **Backend API:** Python, FastAPI (Asynchronous WebSocket streams).
- **AI / NLP:** HuggingFace Transformers, Contextual LLM Orchestration.
- **State & Cache:** Redis (Ephemeral sliding window with 15-min TTL).
- **Frontend / Demo:** React, TypeScript, Tailwind CSS, Recharts.
- **Trust Layer:** Anonymized hash logging for auditable moderation trails.

4. Innovation, Impact & Feasibility

Core Differentiation

- **Proactive vs. Reactive:** Stops abuse during escalation, not post-victimization.
- **Multi-Turn Context:** Detects grooming tactics that appear innocent in single isolated messages.
- **Explainable XAI:** Outputs concrete trigger signals rather than opaque risk percentages.
- **Privacy-First:** Protects children without turning into invasive spyware.

Social & Trust Impact

- **Child Protection:** Shields minors from cyberbullying, online grooming, and social engineering.
- **Digital Trust:** Gives parents and schools confidence in digital learning & gaming spaces.
- **Compliance:** Aligns with COPPA, GDPR-K, and global child online safety regulations.
- **Empowerment:** Teaches digital resilience through timely, contextual guidance.

Implementation Roadmap

- **Hackathon MVP:** Split-screen simulated chat + real-time dynamic risk visualizer.
- **Phase 2:** SDK / API integration for Discord, Roblox, and EdTech platforms.
- **Phase 3:** On-device local SLM execution for zero-cloud latency and complete end-to-end encryption.